

Abinash Koirala

(940) 276-1211 | abinashkoirala23@gmail.com | [LinkedIn](#)

PROFESSIONAL SUMMARY

Data / AI & Machine Learning Engineer with senior-level experience designing and operating scalable data pipelines and cloud data platforms. Led the development of end-to-end ETL/ELT workflows on AWS for semiconductor manufacturing at Texas Instruments, improving data availability for analytics and AI use cases. Built a healthcare data integration solution on Azure at St. David's HealthCare, delivering reliable, governed datasets for clinical and operational reporting. Seeking to apply data architecture and engineering expertise to enable data-driven decision making in a fast-growth organization.

TECHNICAL SKILLS

Programming & Querying: Python, SQL, PySpark, pandas, NumPy, scikit-learn, XGBoost, data manipulation, scripting, automation, complex queries, query optimization, feature engineering

Cloud Platforms & Services: AWS (EMR, S3, Glue, Lambda, Kinesis, Redshift, RDS, EC2, SageMaker, Lake Formation, IAM, CloudWatch), Azure (Azure Data Factory, Azure Databricks, Azure Blob Storage, Azure Purview, Azure Monitor), GCP (BigQuery, Dataproc)

Data Platforms & Warehousing: Snowflake, Amazon Redshift, Azure Synapse, relational databases, operational databases, cloud data warehouses, data marts, analytics-ready datasets

Data Engineering: ETL/ELT, batch data pipelines, data ingestion, data transformation, workflow orchestration, API data integration, data migration, source-to-target mapping, pipeline monitoring

Streaming & Real-Time Technologies: Apache Kafka, Spark Structured Streaming, AWS Kinesis, Azure Event Hubs

AI & Machine Learning: Machine learning-ready datasets, predictive modeling, forecasting, anomaly detection, model-supporting data pipelines, training dataset preparation, feature pipelines, AI-driven analytics, applied ML workflows

Modeling & Data Quality: Data modeling, dimensional modeling, star schema, data validation, reconciliation, data cleansing, data quality auditing, schema drift detection, duplicate detection, outlier detection, data lineage

Tools & Engineering Practices: dbt, Apache Airflow, Git, GitHub, CI/CD fundamentals, automated testing, unit testing, documentation, peer review, troubleshooting, production monitoring, root-cause analysis

Analytics, Domains & Systems: Power BI, Tableau, Excel Power Query, manufacturing analytics, supply chain analytics, inventory analytics, supplier performance, demand planning, healthcare claims, EHR, ERP, Workday, ServiceNow

PROFESSIONAL EXPERIENCE

Texas Instruments

January 2023 - Present

Data / AI & Machine Learning Engineer

Dallas, TX

- Designed, developed, and maintained scalable ETL/ELT pipelines using Python, SQL, PySpark, and cloud-native services to ingest manufacturing, supply chain, ERP, sensor, and operational data into centralized analytics and AI-ready data platforms.
- Built reusable data transformation workflows that standardized raw source data, applied business rules, and produced high-quality datasets for production analytics, inventory optimization, supplier performance, demand planning, and machine learning use cases.
- Developed cloud-based data ingestion and processing solutions using AWS services including Amazon S3, AWS Glue, Amazon Redshift, Amazon Athena, Lambda, and Step Functions to support enterprise reporting, predictive analytics, and downstream AI workloads.
- Created and maintained curated data models in Snowflake and Redshift, improving metric consistency and enabling reliable consumption by Power BI dashboards, business applications, and machine learning feature pipelines.
- Built machine learning-ready datasets by performing feature engineering, data cleansing, aggregation, labeling, and validation to support forecasting, anomaly detection, operational risk analysis, and process optimization initiatives.

- Supported predictive modeling initiatives using Python, pandas, NumPy, scikit-learn, and XGBoost to identify patterns in production trends, inventory movement, supplier delays, and demand fluctuations.
- Assisted in developing AI-driven analytics solutions that combined structured ERP data, operational metrics, and unstructured business inputs to improve decision-making across manufacturing and supply chain teams.
- Built and maintained modular dbt models with automated tests, documentation, and dependency tracking to improve transformation reliability, data lineage, and maintainability of curated enterprise data layers.
- Automated pipeline scheduling, dependencies, retries, alerts, and operational monitoring using Apache Airflow and workflow automation practices, reducing reliance on manual data preparation and improving pipeline reliability.
- Implemented data validation, reconciliation, anomaly detection, and quality checks across ingestion and transformation layers to identify missing records, duplicate data, schema drift, outliers, and processing failures.
- Optimized complex SQL queries, PySpark transformations, and warehouse workloads to improve pipeline performance, reduce processing time, and support timely delivery of operational, analytics, and ML-ready datasets.
- Partnered with engineering, supply chain, finance, analytics, data science, and IT teams to gather requirements, define source-to-target mappings, identify AI/ML use cases, and deliver scalable data solutions.
- Supported Git-based development, peer review, documentation, unit testing, and CI/CD practices for controlled deployment of data pipelines, dbt models, ML-supporting datasets, and reporting layers.
- Monitored production data workflows, investigated pipeline failures and data discrepancies, performed root-cause analysis, and implemented long-term corrective actions to improve data reliability and operational trust.
- Published governed datasets and semantic data layers for Power BI and analytics users, enabling self-service reporting while maintaining consistent KPI definitions, lineage, security, and access standards.
- Collaborated with data science teams to support model development and experimentation by preparing clean training datasets, maintaining feature logic, documenting assumptions, and ensuring data quality across modeling workflows.
- Applied AI-assisted development tools and automation practices to accelerate SQL development, pipeline debugging, documentation, and test case generation while maintaining production-grade engineering standards.
- Contributed to data governance and compliance practices by enforcing access controls, metadata documentation, lineage tracking, and secure handling of sensitive enterprise and operational datasets.

St. David's HealthCare

January 2021 - December 2022

Data Engineer

Austin, TX

- Designed and maintained healthcare data pipelines that ingested patient-service, operational, and clinical data from electronic health systems, relational databases, APIs, and file-based sources.
- Developed ETL/ELT workflows using Azure Data Factory, SQL, and Python to cleanse, transform, and load healthcare data into Azure Synapse Analytics and reporting-ready data layers.
- Built reusable transformation logic and structured data models supporting patient operations, case management, service utilization, workload planning, and compliance reporting.
- Automated scheduled data refreshes and workflow dependencies, improving the reliability and availability of datasets used by clinical, administrative, and operational teams.
- Implemented data quality controls, reconciliation routines, and exception reporting to validate completeness, accuracy, consistency, and integrity across healthcare data sources.
- Troubleshoot pipeline failures, schema changes, source-system inconsistencies, and refresh issues, coordinating with IT and application teams to restore reliable data delivery.
- Integrated data from electronic health records, healthcare information systems, operational databases, and patient-service platforms to create unified datasets for enterprise analytics.
- Developed optimized SQL transformations and Python scripts for data processing, standardization, deduplication, and automation of recurring healthcare data workflows.
- Collaborated with clinical teams, healthcare administrators, IT, and business stakeholders to translate reporting and operational requirements into technical data specifications.
- Supported privacy, confidentiality, access control, and data governance practices for sensitive healthcare data throughout ingestion, processing, storage, and reporting.

- Designed incremental loading and change-handling patterns to process recurring healthcare extracts efficiently while preserving historical records for utilization and operational reporting.
- Maintained technical documentation, data dictionaries, lineage notes, and production runbooks to support audit readiness, knowledge transfer, troubleshooting, and ongoing platform operations.

HCA Healthcare

June 2019 - December 2020

Junior Data Engineer

Houston, TX

- Developed SQL-based ETL processes to extract, transform, validate, and load healthcare claims, billing, utilization, provider, and electronic health record data into reporting databases.
- Built Python and SQL scripts for data cleansing, standardization, reconciliation, and automation of recurring claims and utilization data preparation activities.
- Integrated data from claims databases, billing systems, EHR platforms, payer files, and operational sources to create consolidated datasets for downstream reporting and analysis.
- Performed source-to-target validation, record-count reconciliation, duplicate detection, null checks, and business-rule testing to improve reliability of healthcare data products.
- Created and maintained relational data models and reporting tables that supported claims processing, reimbursement analysis, utilization monitoring, and provider performance reporting.
- Automated scheduled reporting workflows using SQL and Excel Power Query while supporting the transition from manual spreadsheet processes to repeatable data pipelines.
- Investigated data discrepancies, failed loads, payment mismatches, and source-system issues, documenting root causes and coordinating resolutions with technical and business teams.
- Supported secure handling of sensitive healthcare and claims data in alignment with organizational policies, regulatory expectations, and internal data governance standards.
- Collaborated with clinical teams, revenue-cycle management, healthcare administrators, and analysts to gather requirements and deliver reliable data for operational decision-making.
- Wrote reusable stored procedures, views, and complex SQL queries to support batch loads, reporting tables, recurring operational extracts, and downstream analytics requirements.
- Monitored scheduled jobs and data refreshes, escalated failures, and assisted senior engineers with troubleshooting, controlled reruns, defect resolution, and production support activities.
- Maintained data dictionaries, field mappings, transformation specifications, and process documentation to improve traceability, team handoffs, audit support, and long-term maintainability.

EDUCATION

Midwestern State University | *Master of Business Administration, Business Analytics*

Wichita Falls, Texas